

Lab Assignment 19.3 – Code Translation

M.Rushikesh

2303A51399

Task 1: Python to C++ Conversion

Question

Translate a Python class representing a Student into C++.

Prompt

"Convert a Python class with attributes and methods into equivalent C++ code with constructors and proper data types."

Code

Python Code

```
class Student:

    def __init__(self, name, age):

        self.name = name

        self.age = age

    def display(self):

        print(self.name, self.age)
```

```
s = Student("Alice", 20)

s.display()
```

C++ Code

```
#include <iostream>

using namespace std;

class Student {

private:

    string name;
```

```
int age;

public:

    Student(string n, int a) {
        name = n;
        age = a;
    }

    void display() {
        cout << name << " " << age << endl;
    }
};

int main() {
    Student s("Alice", 20);
    s.display();
    return 0;
}
```

Output

Alice 20

Code Explanation

- Python class translated into C++ class
 - Constructor defined using same parameters
 - Private variables used for encapsulation
 - Output remains consistent
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Task 2: Java to Python Conversion

Question

Convert Java isPrime() method into Python.

Prompt

"Translate Java function to check prime number into Python using efficient logic."

Code

Java Code

```
public static boolean isPrime(int n) {  
    if (n <= 1) return false;  
    for (int i = 2; i <= n/2; i++) {  
        if (n % i == 0) return false;  
    }  
    return true;  
}
```

Python Code

```
def is_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, n//2 + 1):  
        if n % i == 0:  
            return False  
    return True  
  
print(is_prime(7))
```

Output

True

Code Explanation

- Java syntax converted to Python syntax
- Loop logic maintained

- Python uses indentation instead of braces
 - Function returns same result
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Task 3: Pseudocode to Python

Question

Convert sorting pseudocode into Python.

Prompt

"Convert bubble sort pseudocode into Python code and test with sample input."

Code

```
def bubble_sort(arr):  
    n = len(arr)  
    for i in range(n):  
        for j in range(0, n-i-1):  
            if arr[j] > arr[j+1]:  
                arr[j], arr[j+1] = arr[j+1], arr[j]  
  
arr = [5, 2, 9, 1]  
bubble_sort(arr)  
print(arr)
```

Output

[1, 2, 5, 9]

Code Explanation

- Implements bubble sort logic
- Swaps adjacent elements
- Iteratively sorts array
- Produces sorted list

Task 4: SQL to Pandas Query

Question

Convert SQL query into Pandas operation.

Prompt

"Convert SQL SELECT query with condition into Pandas DataFrame filtering."

Code

SQL Query

```
SELECT name, salary FROM employees WHERE salary > 50000;
```

Python (Pandas) Code

```
import pandas as pd
```

```
data = {  
    "name": ["Alice", "Bob", "Charlie"],  
    "salary": [60000, 40000, 70000]  
}
```

```
df = pd.DataFrame(data)
```

```
result = df[df["salary"] > 50000][["name", "salary"]]  
print(result)
```

Output

```
   name salary  
0  Alice  60000  
2  Charlie  70000
```

Code Explanation

- SQL WHERE converted to DataFrame filtering
 - Column selection using brackets
 - Same result achieved as SQL query
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Task 5: Algorithm Translation (Real-Time)

Question

Translate a searching algorithm from Python to C.

Prompt

"Convert Python linear search algorithm into C language and compare both implementations."

Code

Python Code

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
print(linear_search([1,2,3,4], 3))
```

C Code

```
#include <stdio.h>  
  
int linear_search(int arr[], int n, int key) {  
    for(int i = 0; i < n; i++) {  
        if(arr[i] == key)  
            return i;  
    }  
    return -1;  
}
```

```
int main() {  
    int arr[] = {1,2,3,4};  
    int result = linear_search(arr, 4, 3);  
    printf("%d", result);  
    return 0;  
}
```

Output

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Code Explanation

- Same logic implemented in both languages
 - Python uses dynamic lists, C uses arrays
 - Memory handling explicit in C
 - Output matches in both implementations
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